

SAMRUDDHI SUTAR

Boston, MA | sutar.s@northeastern.edu | (857) 269 5198 | [LinkedIn](#) | [GitHub](#)

EDUCATION

Northeastern University | Master of Science in Bioinformatics **GPA: 3.9/4**

Boston, MA

Coursework: Data Mining/Machine learning, Statistics in R, Computational Methods.

Aug 2023 - Aug 2025

Ramaiah Institute of Technology | Bachelor of Engineering in Biotechnology **GPA: 3.4/4**

Bangalore, India

Coursework: Biostatistics, Fundamentals of C++, Engineering Mathematics

Aug 2018 - July 2022

EXPERIENCE

Icahn School of Medicine at Mount Sinai

New York, NY

Data Scientist

October 2025 – Present

- Developed an automated AI-agent evaluation pipeline using Python, LLM APIs, and notebook parsing to validate dataset downloads and produce analytics dashboards for multi-run reproducibility.

CytoTronics

Boston, MA

Data Science Intern

Jan 2025 – Aug 2025

- Developed and validated statistical methodologies for analysing 500+ multi-parametric electrophysiological datasets across cardiac, neural, and immuno-oncology applications, supporting translational research for pharmaceutical partners (AstraZeneca, Novartis, AbbVie).
- Designed and implemented reproducible analysis pipelines with integrated QA/QC frameworks, ensuring data quality and consistency across large-scale biological datasets for regulatory-ready deliverables.
- Applied advanced statistical modelling techniques including outlier detection and covariate adjustment strategies to complex longitudinal datasets, improving data integrity by 15% and enabling robust inference in early-phase research settings.
- Designed interactive analytics dashboards using Plotly and custom visualization libraries to deliver 50+ client-facing analytical reports, enabling real-time quality control monitoring, experimental design optimization, and data-driven decision-making.

SigTuple Technologies

Bangalore, India

Research Intern – Microfluidics

Aug 2022 – July 2023

- Engineered end-to-end automated diagnostic pipeline integrating Python-based data acquisition, image processing (OpenCV), and YOLO-based deep learning classification for WBC differential counting across 7 cell types, processing 1,000+ blood samples.
- Co-inventor on patent application for novel alcohol-free staining reagent formulation, designing and executing statistical validation studies using DOE (Design of Experiments) and regression analysis to optimize reagent performance across variable sample conditions, reducing assay time by 50% while maintaining clinical accuracy standards.

Stellixir Biotech

Bangalore, India

Research Intern

Aug 2021 – Dec 2021

- Designed and executed in vitro cancer studies following standardized protocols (MTT, CCK-8, cell counting assays), ensuring data integrity and reproducibility for translational research applications through rigorous experimental controls and statistical validation of biological replicates.
- Performed statistical analysis and visualization of cytotoxicity assay results using R and Python (ggplot2, Matplotlib), integrating findings into automated computational pipelines to support data-driven experimental decision-making.

SKILLS AND CERTIFICATIONS

Programming & Statistics: Python, R, SQL | Regression Analysis, Hypothesis Testing, Statistical Modeling, Sequence Analysis

Python Libraries: Pandas, NumPy, Matplotlib, Seaborn, Plotly, OpenCV, scikit-learn, PyTorch, TensorFlow, BioPython, Dask

Data & Visualization: Tableau, PowerBI, Plotly, ggplot2, Interactive dashboards

Infrastructure & DevOps: Unix/Linux Shell, Git/GitHub, Docker, AWS, Jupyter Notebooks, VSCode, High-performance computing

Certifications: Coursera - Computer Programming for Biologists- R and Python, Python for Genomic Data Science, Understanding Cancer Metastasis (John Hopkins University), Biomedical Nanotechnology (Indian Institute of Technology-Roorkie)

PROJECTS AND PUBLICATIONS

- Automated Tomato Quality Classification using Deep Learning:** Built real-time computer vision pipeline using YOLOv5 processing 30-60 FPS with 93.5% mAP on imbalanced dataset through transfer learning and data augmentation.
- Heart Failure Mortality Prediction Using Ensemble Methods:** Built ensemble ML model (Decision Tree, Logistic Regression, SVM) predicting patient mortality with 83.3% accuracy and 90% sensitivity using 13 clinical biomarkers.
- Comprehensive bioinformatics analysis of structural and functional consequences of deleterious missense mutations in the human QDPR gene. DOI: [10.1080/07391102.2023.2226740](https://doi.org/10.1080/07391102.2023.2226740)
- A Computational Study on Structural and Functional Consequences of nsSNPs in Human Dopa Decarboxylase. DOI: [10.1080/07391102.2023.2301517](https://doi.org/10.1080/07391102.2023.2301517)